Phylogenomics and time-calibrated analyses of cyclostome divergence, diversity and biogeography

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Cyclostomes

- The only extant lineage of jawless vertebrates (Agnatha)
- Comprising lampreys (Petromyzontida) and hagfishes (Myxini)
- Once considered paraphyletic, now strongly supported as monophyletic by molecular data
- Phylogenomic analyses of relationships, biogeography, and diversification timing remain limited—especially for hagfishes[1,2]
- The origin of Galápagos hagfish diversity remains unresolved[3]
- **Here, we use phylogenomics and biogeographic inference to test hypotheses of diversification and colonization in Galápagos hagfishes**

Methods

Data collection
- RNA-seq data generated for three hagfish genera (Myxine, Eptatretus, Neomyxine) using Illumina NovaSeq 6000
- Additional genomic, transcriptomic, and mitochondrial (16S, COX1) data obtained from GenBank[i]
- Taxonomic and geographic range data compiled from literature and FishBase[5]
- Genome and transcriptome assemblies generated using Trinity[ii], MEGAHIT[iii], and SPAdes[iv]
- Missing mitochondrial markers recovered from RNA-seq data using BLAST[v]

Ortholog identification & filtering
- Single-copy orthologs identified across 50 species using BUSCO[6]
- Sequences aligned with MAFFT[7] and trimmed with trimAl[8]
- Loci present in <20 species excluded
- Loci with compositional heterogeneity removed using IQ-TREE[9]

Locus selection
- Phylogenetic signal assessed using GeneSortR[10] (loci ranked by signal strength and consistency)
- Top 150 loci retained for downstream analyses

Gene tree inference & curation
- Gene trees inferred for each locus using IQ-TREE[11]
- Spurious sequences and long-branch outliers removed using TreeShrink[12]

Phylogenetic reconstruction
(Fig. 2)
- Partitioned analysis of 150 nuclear loci and mitochondrial genes (16S, COX1) performed in IQ-TREE[11]
- Node support assessed using ultrafast bootstrap, SH-aLRT, and gene concordance factors (gCF)

Divergence time estimation
(Fig. 3)
- Calibration points compiled from previous studies via TimeTree[13]
- Time-calibrated phylogeny inferred using MCMCTree[14]

Biogeographic inference
(Fig. 3 & 4)
- Outgroups pruned from the time-calibrated phylogeny
- Geographic regions defined using realms from Marine Ecoregions of the World[15]
- Ancestral ranges inferred using BioGeoBEARS[16]

Figure 2. Analytical workflow of this study.

Cyclostome time-tree reveals an ancient split, but more recent diversification of extant hagfish and lamprey diversity

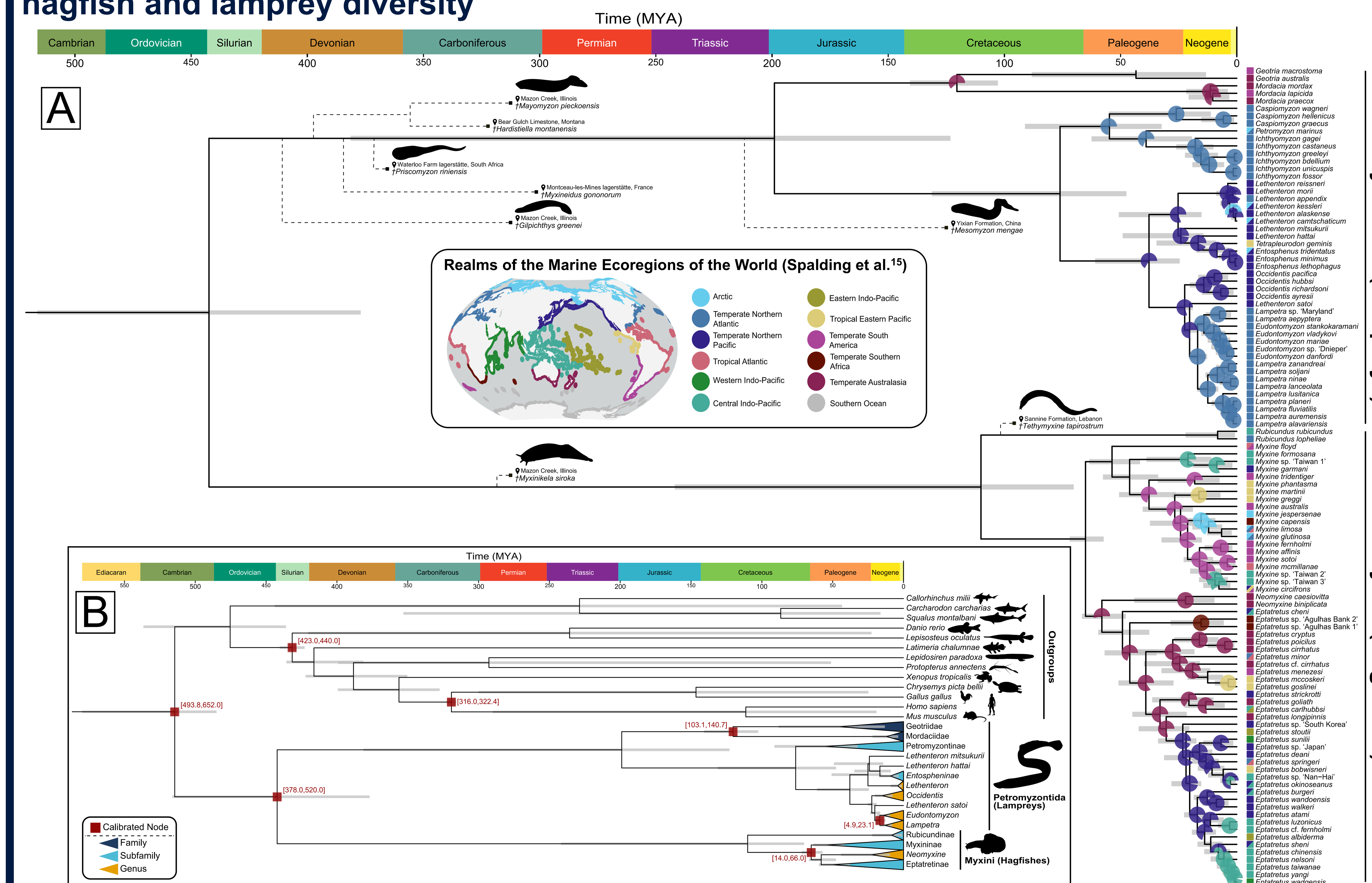

Figure 3. Images throughout the figure are either public domain (PhyloPic) or adapted from CC-licensed sources. **(A)** Time-calibrated phylogeny of cyclostomes (the focal group for biogeographic analyses) with pie charts showing inferred ancestral geographic ranges. Multiple colors within pies represent the full reconstructed range, and pie completeness reflects reconstruction confidence. Node bars indicate 95% confidence intervals; ranges extending beyond axis breaks are shown in brackets. Colored boxes preceding taxon names indicate present-day ranges; for freshwater and anadromous species, ranges are assigned to the nearest marine ecoregion at river outlets. Black squares (■) indicate estimated timing of fossil cyclostomes, and dashed lines indicate inferred topology only (not divergence timing). Fossil age and phylogenetic placement are from Miyashita *et al.*[17]. **(B)** Full time-calibrated phylogeny including outgroups; monophyletic groups are collapsed for clarity.

Phylogenomic analysis clarifies cyclostome relationships

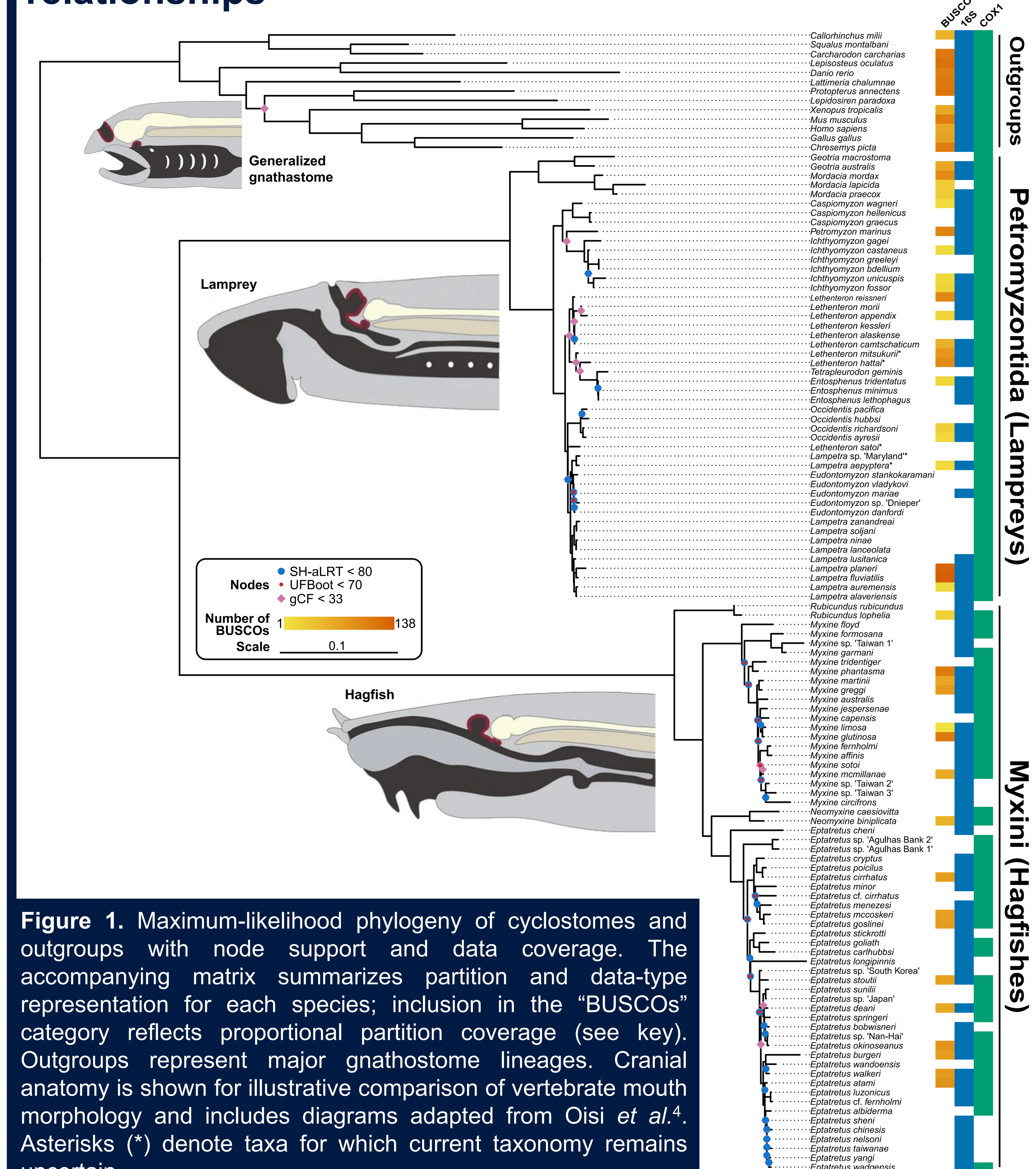

Figure 1. Maximum-likelihood phylogeny of cyclostomes and outgroups with node support and data coverage. The accompanying matrix summarizes partition and data-type representation for each species; inclusion in the "BUSCOs" category reflects proportional partition coverage (see key). Outgroups represent major gnathostome lineages. Cranial anatomy is shown for illustrative comparison of vertebrate mouth morphology and includes diagrams adapted from Oisi *et al.*[4]. Asterisks (*) denote taxa for which current taxonomy remains uncertain.

Results

Phylogenomic relationships
(Fig. 1)
- Cyclostomes form a strongly supported clade
- Deep relationships among major lineages are well resolved
- Some intrageneric nodes show lower support

Divergence times
(Fig. 3)
- Cyclostomes exhibit an ancient hagfish–lamprey split
- Most extant diversity is concentrated in recent radiations

Biogeography
(Fig. 3 & 4)
- Lineages show structured geographic histories over time

Discussion

- Cyclostomes are strongly supported as a monophyletic clade.
- Hagfish genera are largely monophyletic, while lamprey relationships are generally well supported but retain limited generic-level uncertainty; under genomic data, *Neomyxine* is recovered as sister to *Eptatretus*, contrasting with previous interpretations.
- Molecular dating supports an ancient hagfish–lamprey split (~430 MYA), with most extant diversity arising within the last ~100 MY.
- Both hagfish and lampreys exhibit strong biogeographic structuring, including the Galápagos hagfishes, which reflect multiple independent colonization events rather than a single regional origin.

Biogeographic analysis indicates multiple colonizations in Galápagos hagfish

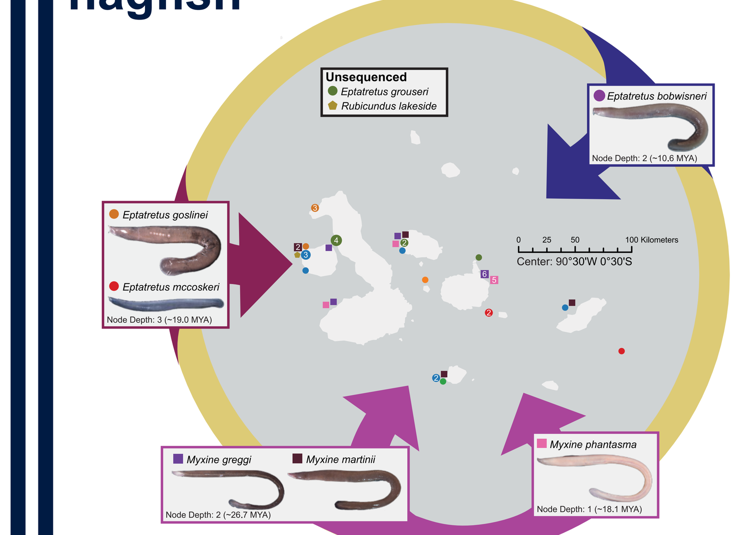

Figure 4. Map of hagfish observations in the Galápagos since 1980[18], with predicted ancestral range reconstructions overlaid. Node depth indicates the number of nodes from the tips (extant species) at which ancestral ranges are inferred (see Fig. 3).

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